Pilot experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Preserve pilot defects and revisions while keeping new primary evidence separate.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

Predefining an analysis plan

My remembered idea, example and correction

Research ethics and consent

My remembered idea, example and correction

Ablation studies

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A pilot checks feasibility, capture, units, rubric and analysis before a primary study. Main five-case pilot reveals wrong duration-log units. Preserve the original defect and correction; without actual values or unit definitions, no numerical conversion can be calculated. A method revised using pilot results needs newly reserved primary cases for an untouched final comparison.

  Separate picture: two reviewers score four practice answers, agree on three and disagree on one. Observed agreement is 3/4 = 75%; disagreement is 25%. No reference labels or disagreement cause are supplied. Agreement does not prove correctness or long-run reliability. Inspect scoring records and clarify the rubric, retaining the pilot as development evidence.

  Adding corrected practice cases to final evaluation would mix roles under the stated plan. Preserve versions, test the repair on appropriate fixtures and fix the revised analysis plan before new primary outcomes. No actual timing run, reviewer study or final effect estimate is supplied.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- **Pilot** five cases reveals wrong units in duration logs; team corrects units and reports the **pilot** as final **evidence**. Show the intermediate steps and give the conclusion with its scope.

Main pilot step	Supported action	Unsupported claim
Five-case pilot exposes wrong duration units	Document defect and affected measurement	Pretend the original logs were correct
Correct unit handling	Record expected units, actual units and conversion evidence	Invent a numerical conversion without the missing values
Method revised using pilot	Retain pilot as development evidence	Call revised pilot outcomes an untouched final comparison
Freeze revised procedure	Record new version and planned measurement boundaries	Claim version label alone proves chronology

Main pilot step	Supported action	Unsupported claim
New primary cases	Reserve and evaluate under the fixed revised plan	Claim primary results before running the study

> **My reasoning and evidence**
>
> _____
>
> _____
>
> _____
>
> _____

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A pilot checks feasibility, capture, units, rubric and analysis before a primary study. Main five-case pilot reveals wrong duration-log units. Preserve the original defect and correction; without actual values or unit definitions, no numerical conversion can be calculated. A method revised using pilot results needs newly reserved primary cases for an untouched final comparison.

  Separate picture: two reviewers score four practice answers, agree on three and disagree on one. Observed agreement is 3/4 = 75%; disagreement is 25%. No reference labels or disagreement cause are supplied. Agreement does not prove correctness or long-run reliability. Inspect scoring records and clarify the rubric, retaining the pilot as development evidence.

  Adding corrected practice cases to final evaluation would mix roles under the stated plan. Preserve versions, test the repair on appropriate fixtures and fix the revised analysis plan before new primary outcomes. No actual timing run, reviewer study or final effect estimate is supplied.

- A report omits this qualification: A **pilot** may assess **feasibility** but is not automatically powered or independent **evidence** for the final hypothesis.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A pilot checks feasibility, capture, units, rubric and analysis before a primary study. Main five-case pilot reveals wrong duration-log units. Preserve the original defect and correction; without actual values or unit definitions, no numerical conversion can be calculated. A method revised using pilot results needs newly reserved primary cases for an untouched final comparison.

  Separate picture: two reviewers score four practice answers, agree on three and disagree on one. Observed agreement is 3/4 = 75%; disagreement is 25%. No reference labels or disagreement cause are supplied. Agreement does not prove correctness or long-run reliability. Inspect scoring records and clarify the rubric, retaining the pilot as development evidence.

  Adding corrected practice cases to final evaluation would mix roles under the stated plan. Preserve versions, test the repair on appropriate fixtures and fix the revised analysis plan before new primary outcomes. No actual timing run, reviewer study or final effect estimate is supplied.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- **Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **pilot** experiments **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **pilot** experiments.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Pilot** experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Pilot** experiments. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Pilot** six cases exposes ambiguous labels in two. What should happen before the primary study?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

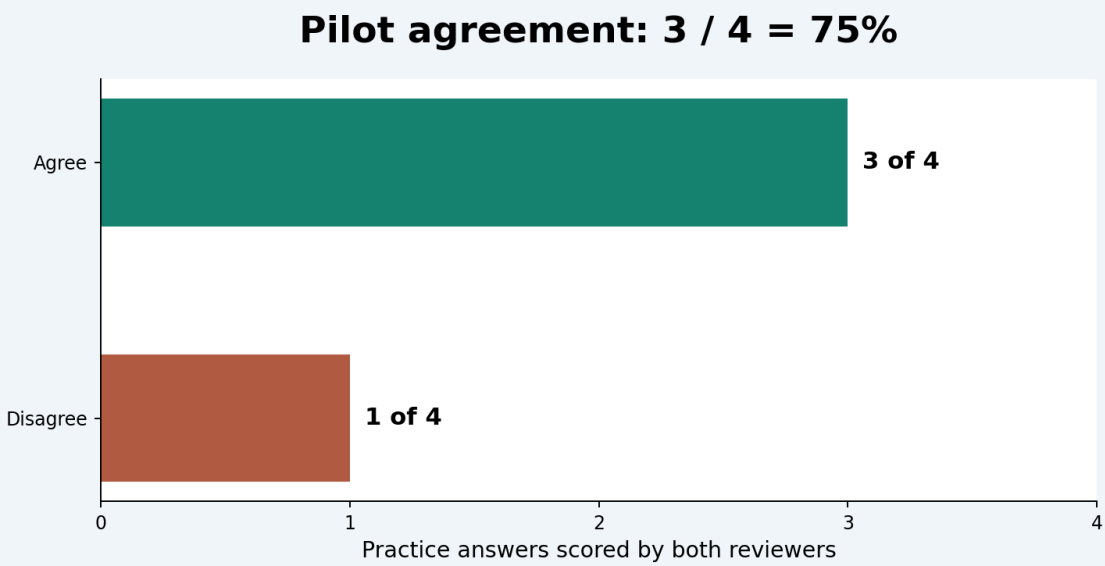
> Keep this task's inputs,rule and response together.

Complete task and evidence

- A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.
- Two reviewers score four practice answers. They agree on three and disagree on one; these answers are excluded from final evaluation.
- Calculate reviewer agreement on the four **pilot** answers. What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

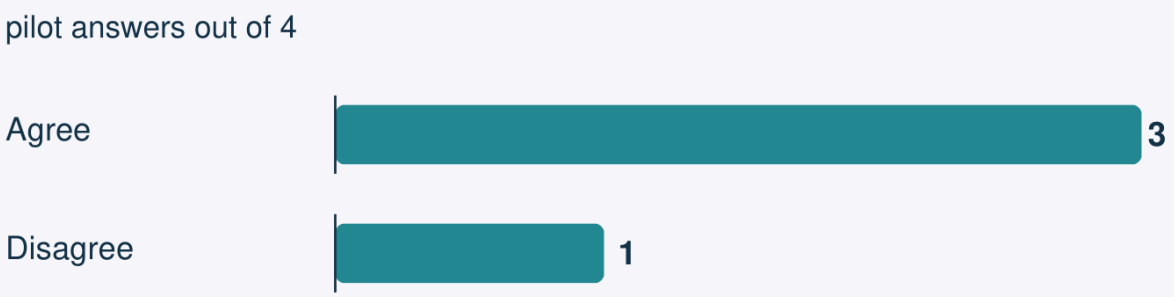
  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



One disagreement can reveal a rubric problem; its cause is not supplied.
These four revised practice answers are development evidence, not untouched final cases.
Agreement is not correctness or a long-run reliability estimate.

AI ACADEMY / AI-7/22

Reviewer agreement on four pilot answers



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/22

Pilot experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Try the workflow

Both reviewers apply the draft rubric to four practice answers.



2. Measure agreement

Agreement is 3/4=75% on this pilot.



3. Repair the rubric

Discuss the disagreement and clarify the scoring rule before the main run.

CHANGE ONE THING

Add the corrected pilot answers into final evaluation.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study.
- Which labelled step matches this detail? Discuss the disagreement and clarify the scoring rule before the main run.
- 1. Try the workflow
- 2. Repair the rubric
- 3. Measure agreement
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Repair the rubric Discuss the disagreement and clarify the scoring rule before the main run.

A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Changed-case question and optional hint

- Changed-case practice: Add the corrected **pilot** answers into final evaluation. Which conclusion is supported by the supplied facts?
- 1. Declare them previously exposed rather than untouched final **evidence**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Declare them previously exposed rather than untouched final **evidence**. Rubric development already used those answers.

Repair and retain the record

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

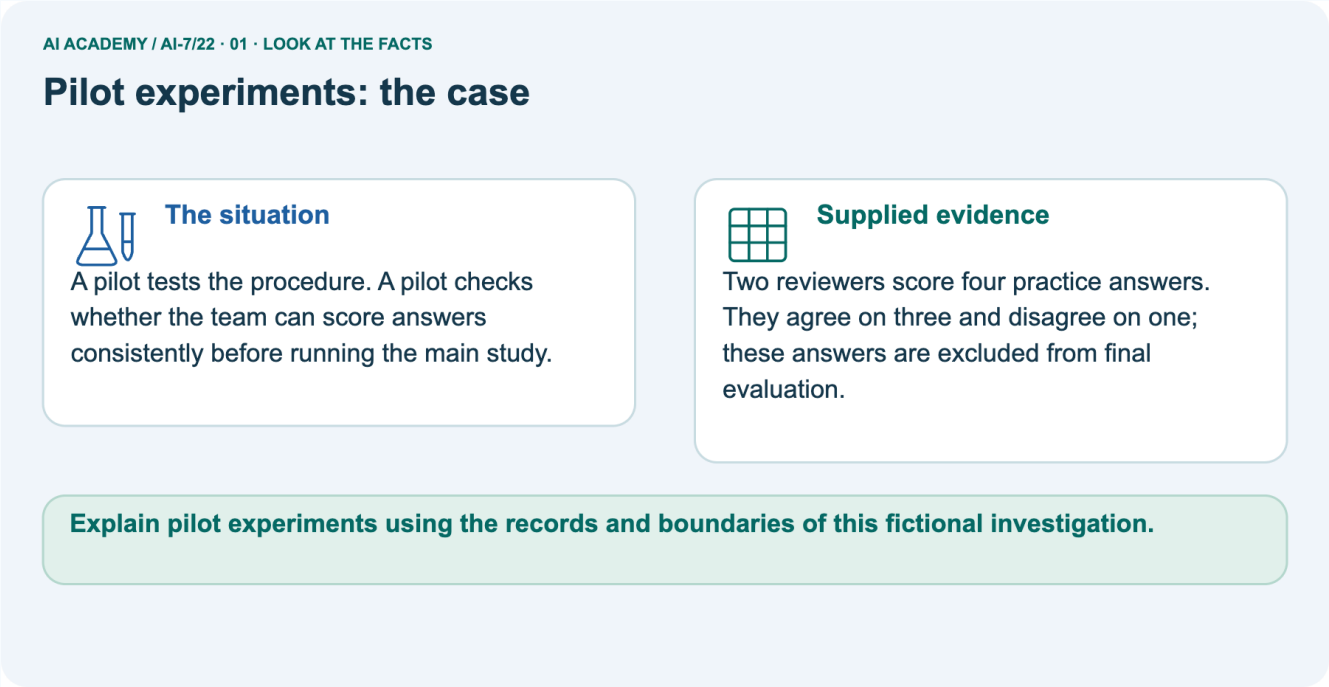
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Pilot experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

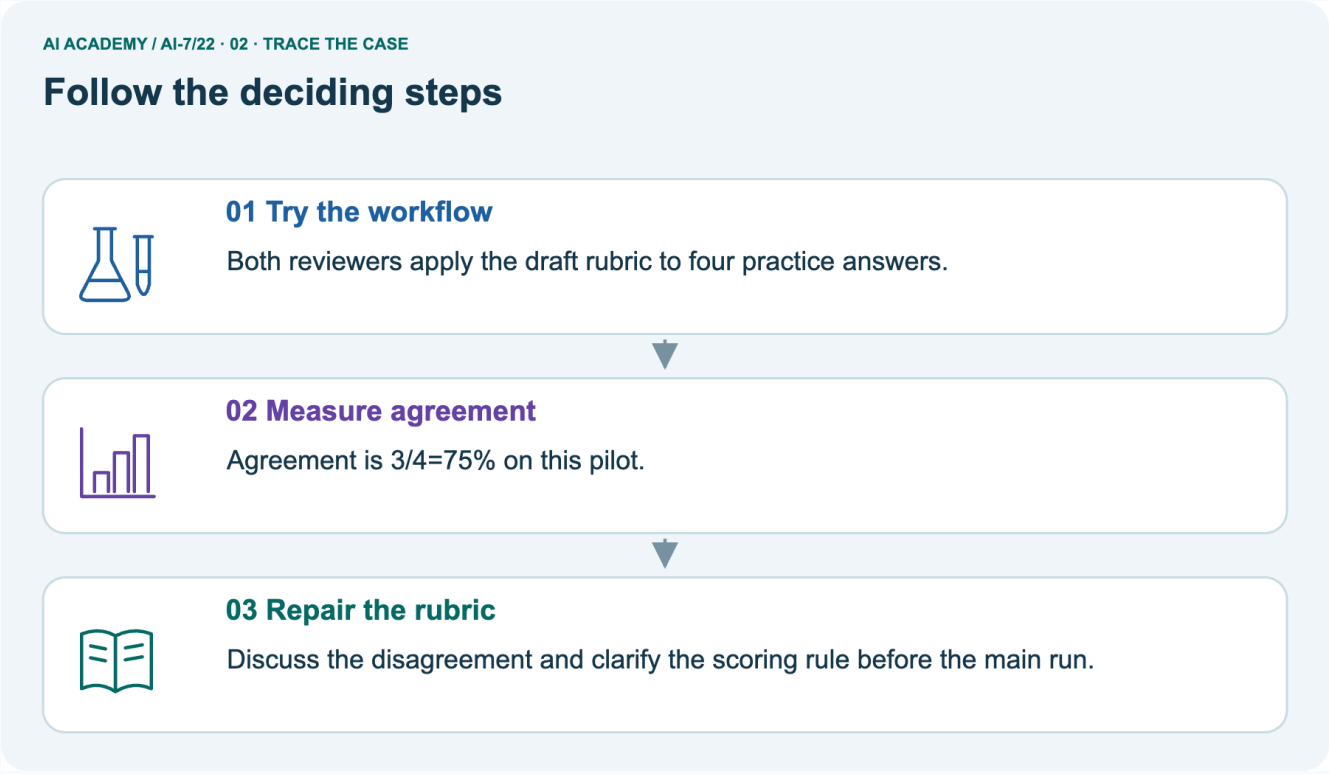
A **pilot** tests the procedure. A **pilot** checks whether the team can score answers consistently before running the main study. Supplied facts: Two reviewers score four practice answers.

They agree on three and disagree on one; these answers are excluded from final evaluation.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



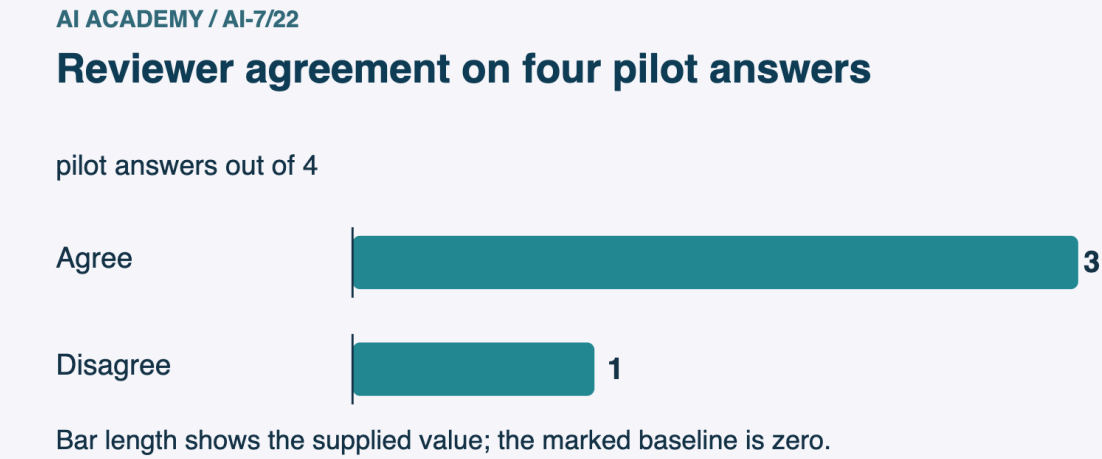
Follow the labelled picture with the matching explanation. Open the vector diagram

Try the workflow: Both reviewers apply the draft rubric to four practice answers. → Measure agreement: Agreement is 3/4=75% on this **pilot**. → Repair the rubric: Discuss the disagreement and clarify the scoring rule before the main run.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Reviewer agreement on four pilot answers



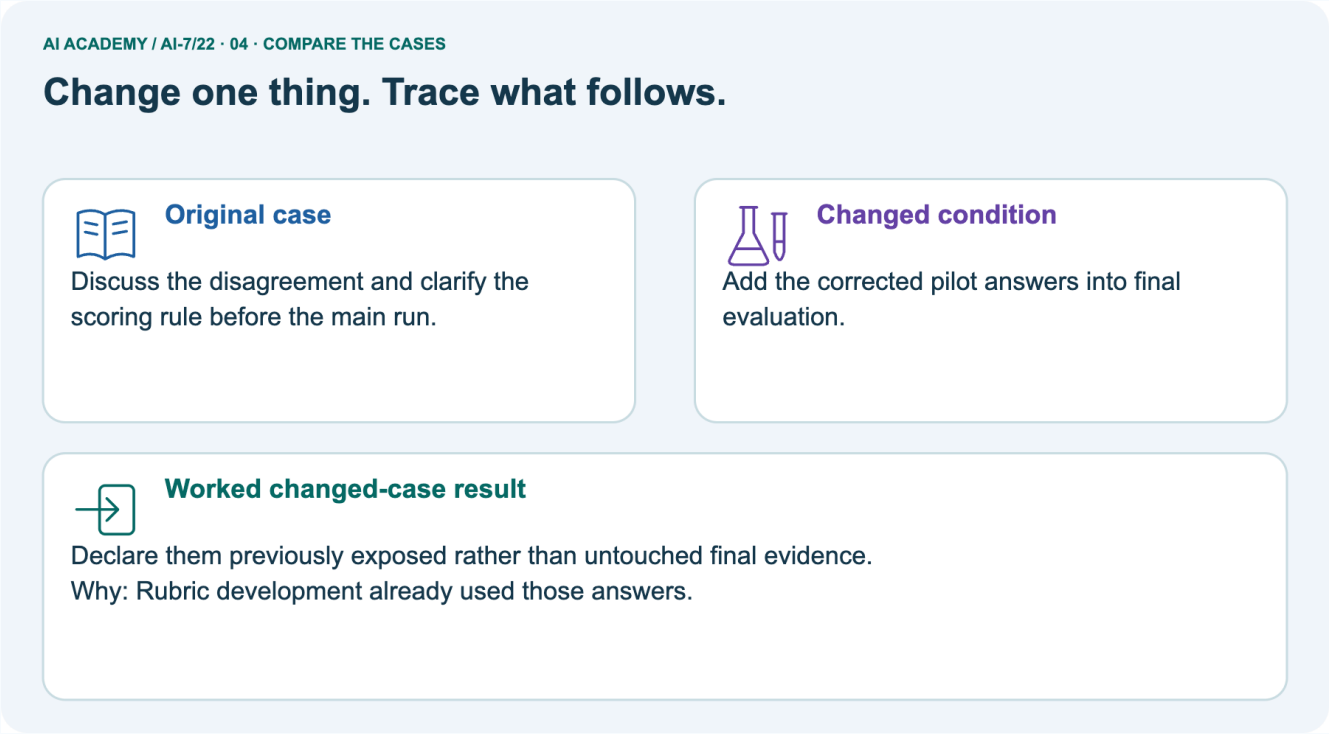
Follow the labelled picture with the matching explanation. Open the vector diagram

Reviewer agreement on four **pilot** answers. Units: **pilot** answers out of 4. Agree: 3. Disagree: 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

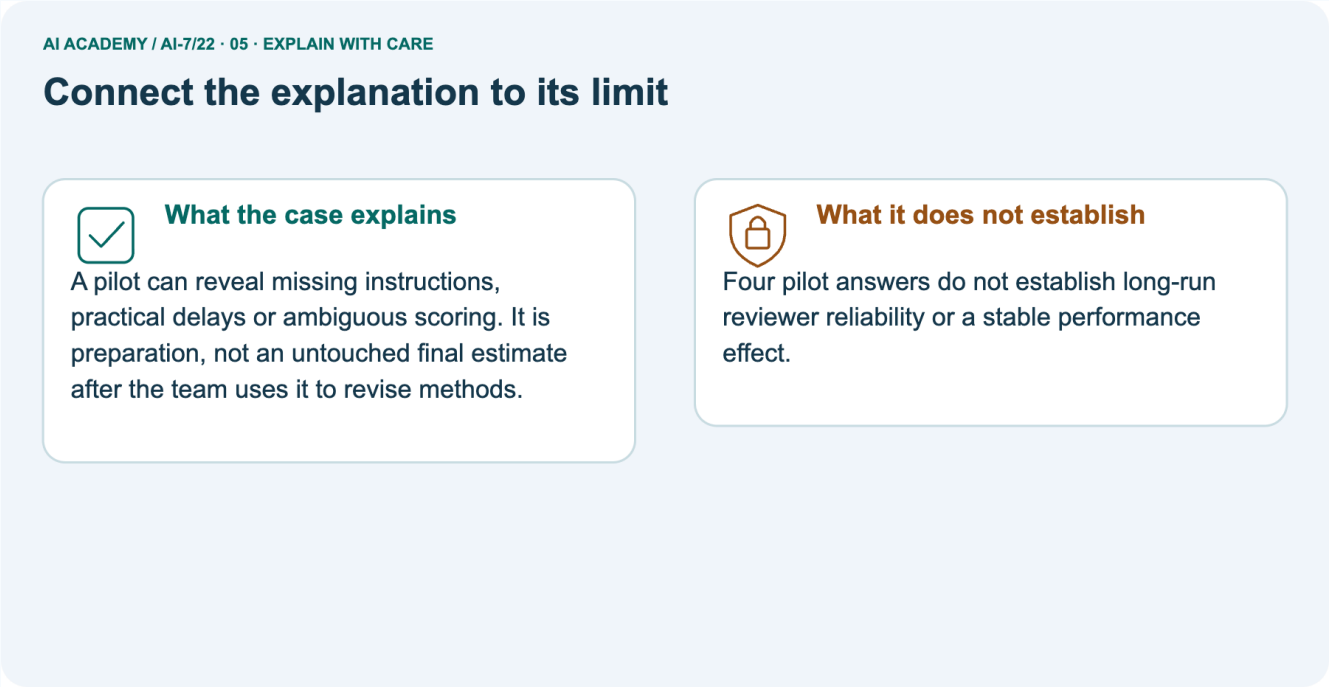
Original case: Discuss the disagreement and clarify the scoring rule before the main run.
Changed condition: Add the corrected **pilot** answers into final evaluation.

Expected result: Declare them previously exposed rather than untouched final **evidence**.
Reason: Rubric development already used those answers.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

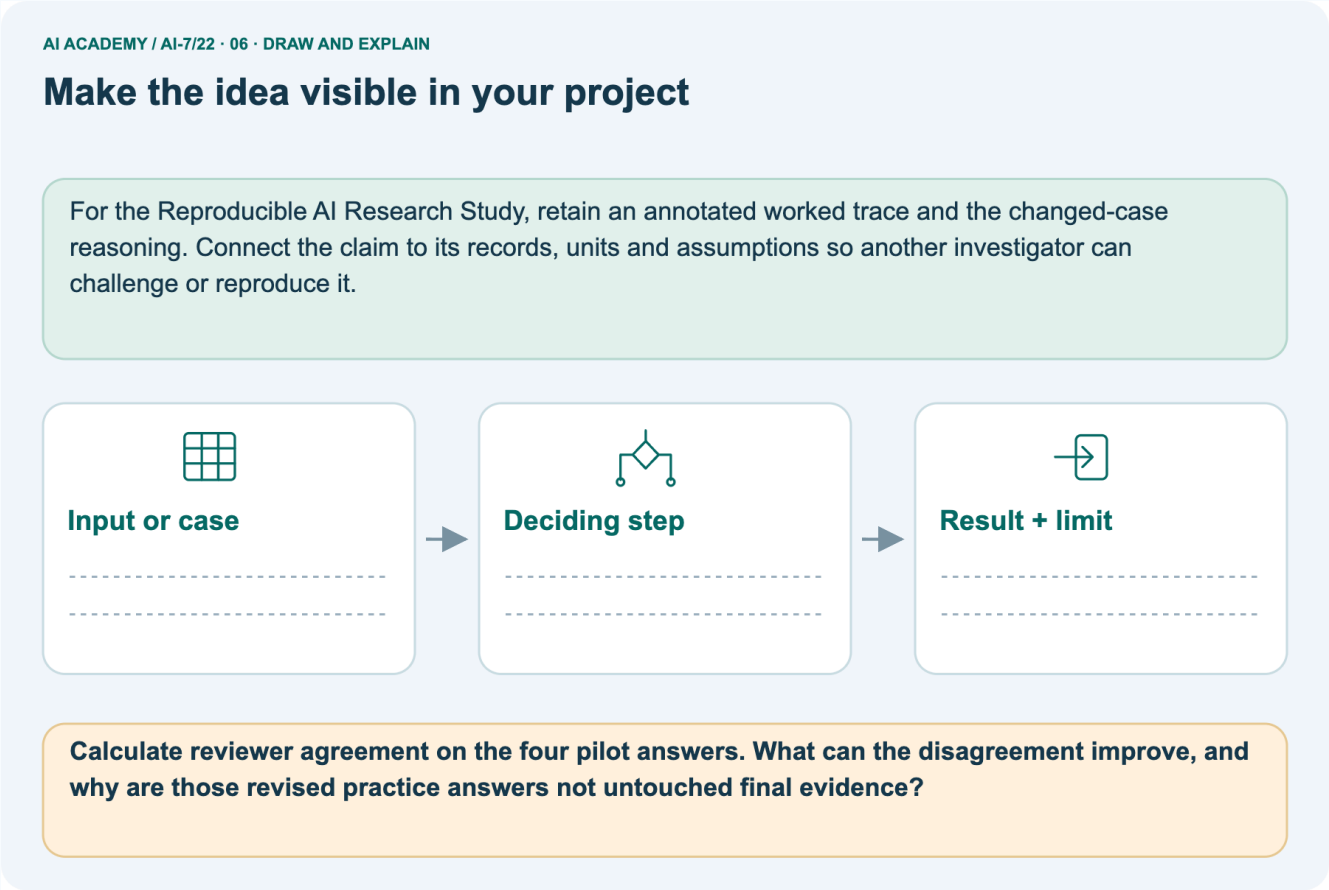
A **pilot** can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Limit: Four **pilot** answers do not establish long-run reviewer reliability or a stable performance effect.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate reviewer agreement on the four **pilot** answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Pilot experiments: the case

02. Follow the deciding steps

03. Reviewer agreement on four pilot answers

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.